



Stage 3: Algorithmic Thinking — LeetCode Practice Alignment

Legend

Stage 3 of the 5-Stage Placement Training Framework: Algorithmic Thinking. Bridges Stage 1 (Syntax — Python/Java Core Examples) and Stage 2 (Must-Know Algorithms — Topic-Wise Question Banks) into applied LeetCode practice. Each list is aligned to the corresponding Stage 2 topic's core concepts only. Problems requiring techniques not yet covered in Stage 4/5 (tree/graph traversal, DP, two-pointer, recursion-heavy patterns) are deliberately excluded. OOPs, Exception Handling, Streams & Lambda, and Multithreading are excluded from this guide — LeetCode has little to no genuine alignment for those topics.

1. Conditional Statements

- LC 412 — Fizz Buzz
- LC 1108 — Defanging an IP Address
- LC 728 — Self Dividing Numbers
- LC 258 — Add Digits
- LC 9 — Palindrome Number
- LC 976 — Largest Perimeter Triangle

2. Control Statements (Loops)

- LC 202 — Happy Number
- LC 507 — Perfect Number
- LC 263 — Ugly Number
- LC 231 / 326 / 342 — Power of Two / Three / Four
- LC 509 — Fibonacci Number
- LC 172 — Factorial Trailing Zeroes

3. Arrays

- LC 66 — Plus One
- LC 26 — Remove Duplicates from Sorted Array
- LC 88 — Merge Sorted Array
- LC 189 — Rotate Array
- LC 283 — Move Zeroes
- LC 448 — Find All Numbers Disappeared in an Array
- LC 1732 — Find the Highest Altitude
- LC 1470 — Shuffle the Array

4. Strings



- LC 344 — Reverse String
- LC 125 — Valid Palindrome
- LC 242 — Valid Anagram
- LC 14 — Longest Common Prefix
- LC 58 — Length of Last Word
- LC 383 — Ransom Note
- LC 796 — Rotate String
- LC 917 — Reverse Only Letters

5. List (ArrayList & LinkedList)

- LC 1929 — Concatenation of Array
- LC 1051 — Height Checker
- LC 1480 — Running Sum of 1d Array
- LC 1913 — Maximum Product Difference Between Two Pairs

Classic LinkedList algorithms (reverse a list, cycle detection, merge two sorted lists) are intentionally excluded — these belong to the DSA roadmap's dedicated Linked List phase.

6. Set

- LC 217 — Contains Duplicate
- LC 705 — Design HashSet
- LC 349 — Intersection of Two Arrays
- LC 1207 — Unique Number of Occurrences
- LC 1748 — Sum of Unique Elements

7. Map

- LC 1 — Two Sum
- LC 706 — Design HashMap
- LC 387 — First Unique Character in a String
- LC 1512 — Number of Good Pairs
- LC 1832 — Check if the Sentence Is Pangram

8. Queue & Deque

- LC 933 — Number of Recent Calls
- LC 622 — Design Circular Queue
- LC 641 — Design Circular Deque
- LC 1046 — Last Stone Weight
- LC 703 — Kth Largest Element in a Stream

9. Stack



- LC 20 — Valid Parentheses
- LC 155 — Min Stack
- LC 232 — Implement Queue using Stacks
- LC 225 — Implement Stack using Queues
- LC 150 — Evaluate Reverse Polish Notation
- LC 1047 — Remove All Adjacent Duplicates In String
- LC 682 — Baseball Game

10. Collections Utility Class

- LC 1122 — Relative Sort Array
- LC 1636 — Sort Array by Increasing Frequency

11. Searching and Sorting

Searching

- LC 704 — Binary Search
- LC 35 — Search Insert Position
- LC 278 — First Bad Version
- LC 374 — Guess Number Higher or Lower
- LC 852 — Peak Index in a Mountain Array

Sorting

- LC 912 — Sort an Array
- LC 977 — Squares of a Sorted Array
- LC 561 — Array Partition
- LC 1287 — Element Appearing More Than 25% in a Sorted Array

This content maps to the Arrays file, Sections 3 (Searching) and 4 (Sorting). No standalone Searching & Sorting file currently exists in the 14-file set.